

Uncovering Trends in Police Shootings: A Data-Driven Dashboard

In recent years, there has been heightened public interest in understanding the nature, frequency, and contextual circumstances of police-involved shootings in the United States. Our project seeks to create a descriptive dashboard that surfaces patterns and insights across time, geography, demographics, and incident characteristics. We will primarily use the dataset from Kaggle which encompasses variables such as date, location (city, state), victim demographics (age, gender, race), armed status, threat level, whether the shooting was fatal, and other contextual factors. By exploring this rich set of attributes, the dashboard will enable users to answer foundational questions of what happened, where, and to whom.

The second part of the project focuses on structuring the data story into clear, visually-driven layers. First, we will present certain key metrics (e.g., total number of shootings, percentage fatal, breakdown by year) to establish the baseline. Next, we will incorporate temporal analytics: trends over years, monthly or seasonal fluctuations, and changes in victim demographics over time. Third, we will map geographic patterns, such as state- and city-level incidence rates, hotspot identification, and regional comparisons. Additional slices will include victim characteristics (race, gender, age), incident context (armed/unarmed status, type of threat), and comparative analysis across demographic groups. These interactive visuals will provide users with the ability to filter and drill down to specific incident types.

Ultimately, the descriptive dashboard aims not to predict future events or attribute causation, but rather to illuminate the landscape of police-involved shootings through clear, accessible storytelling of the data. By doing so, the dashboard supports analysts, researchers, and the public in developing a stronger, evidence-based understanding of incident patterns—highlighting which states and demographic groups are most impacted, how trends have changed over time, and where further investigation may be warranted. This foundational insight will serve as a springboard for further research and decision-making around policy, training, oversight, and community engagement.

Data Source: <https://www.kaggle.com/datasets/ahsen1330/us-police-shootings>

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